

Title: SCALES, RELIEF, AND CONTOURS

National Cadet Corps (NCC) â Official Directorate General Training Notes

1. DEFINITION OF SCALE & REPRESENTATIVE FRACTION (RF): Scale expresses the fixed mathematical ratio between a distance measured on the map and the corresponding distance on the ground. RF 1:50,000 means 1 unit on the map equals 50,000 identical units on the terrain.

2. RELIEF & CONTOUR LINES: Relief represents the three-dimensional elevation variations of the earth's surface. Contours are brown lines connecting points of equal height above Mean Sea Level (MSL). The vertical distance between adjacent contour lines is termed the Vertical Interval (VI), standardly 20 meters.

3. TERRAIN FEATURE IDENTIFICATION: Contour lines positioned close together indicate a steep slope or cliff face. Contour lines widely spaced signify a gentle, flat plain. Concentric circular contours with height numbers increasing toward the center indicate a Hill Peak, while concentric contours with numbers decreasing inward denote a Crater or Basin.

4. SPURS, VALLEYS, AND SADDLES: A V-shaped contour pointing down towards lower elevation represents a Spur (ridge). A V-shaped contour pointing up towards higher elevation represents a Valley or Re-entrant. A dip between two adjacent peaks is a Saddle or Col.